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Python Interview Series-1

PART 1 — STRING FUNDAMENTALS

1. Reverse a string without using built-in functions.
2. Check if a string is a palindrome.
3. Count vowels and consonants in a string.
4. Convert string to title case without using `.title()`.
5. Remove all spaces from a string.
6. Count occurrences of each character in a string.
7. Find the first non-repeating character.
8. Find the most frequent character in a string.
9. Check if two strings are anagrams.
10. Remove duplicate characters while preserving order.

PART 2 — STRING PROBLEM SOLVING

11. Find all substrings of a string.
12. Check if one string is a rotation of another.
13. Find the longest word in a sentence.
14. Count words in a sentence.
15. Reverse the words in a sentence.
16. Check if a string contains only digits.
17. Replace characters without using `.replace()`.
18. Find the longest substring without repeating characters.
19. Compress a string using run-length encoding (e.g., `aaabb -> a3b2`).
20. Check if two strings differ by exactly one character.

PART 3 — LIST FUNDAMENTALS

21. Reverse a list.
22. Find the largest element in a list.
23. Find the second largest element in a list.
24. Remove duplicates from a list.
25. Find common elements between two lists.

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26. Count frequency of elements in a list.
27. Check if a list is sorted.
28. Rotate a list by k positions.
29. Find a missing number in a sequence list.
30. Merge two sorted lists.

PART 4 — LIST ALGORITHMS

31. Flatten a nested list.
32. Find pairs in a list with a given sum.
33. Solve the Two Sum problem.
34. Implement sliding window maximum.
35. Find the majority element in a list.
36. Find duplicates in a list.
37. Find intersection of three lists.
38. Find a subarray with a given sum.
39. Implement a stack using a list.
40. Implement a queue using a list.

PART 5 — TUPLE CONCEPTS

41. Create a tuple and access its elements.
42. Perform tuple unpacking.
43. Swap two variables using tuple unpacking.
44. Convert a list to a tuple.
45. Convert a tuple to a list.
46. Count occurrences of an element in a tuple.
47. Find the index of an element in a tuple.
48. Loop through a tuple using for and enumerate.
49. Use a tuple as a dictionary key.
50. Iterate through nested tuples.